

Spatial gradients in ecohydrologic properties within a pinyon-juniper ecosystem

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ABSTRACT

The influence of woody vegetation and biological soil crusts on infiltration capacity is one of the several uncertainties associated with the ecohydrologic effects of woody plant encroachment into arid and semi-arid land systems. The objective of this study was to quantify the effects of Utah juniper (*Juniperus osteosperma*) and pinyon pine (*Pinus edulis*) on subcanopy and intercanopy ecohydrologic properties. We measured soil sorptivity, unsaturated hydraulic conductivity [K(h)], soil water content (SWC), and water repellency along radial line transects from under *Juniperus* and *Pinus* trees into the centre of the intercanopy space between trees. In the subcanopy, litter mounds, hydrophobic soils, and roots all appear to contribute to preferential flow to below-surface soils via wetted patches. For both *Juniperus* and *Pinus*, K(h) was significantly lower in the subcanopy than the intercanopy average; however, line transect measurements did not show distinct boundaries in K(h) between the two regions. K(h) increased by eight-fold across a gradient extending outward from near the edge of the canopy to approximately 2 times the canopy radius (CR). This suggests that the influence of these species on hydrologic properties extends significantly into the intercanopy region. Analysis of biological soil crust cover within the intercanopy showed that beyond the gradient zone, increasing structural development of biological soil crust was associated with increased K(h). Furthermore, these results indicate that the distance from the canopy and direction should be considered in the assessment and modelling of woody plant and biological soil crust influence on infiltration capacity. Copyright © 2008 John Wiley & Sons, Ltd.

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INTRODUCTION

Management of rangelands requires an understanding of ecosystem controls on specific resources, and ecosystem response to management actions. The development of hydrophobic soil has been shown to be a key process affecting hydrologic properties of many woody species (Scholl, 1971; Doerr *et al.*, 2000; Pierson *et al.*, 2003; Mataix-Solera *et al.*, 2007). However, despite this acknowledgement, detailed analyses of its effects are rarely considered in typical non-fire related studies in the USA.

Juniper (*Juniperus*) and Pinyon (*Pinus*) trees are examples of plant species, that have been documented in modifying the soil surface through the development of a hydrophobic layer (i.e. Jaramillo *et al.*, 2000). Woodlands dominated by these species are of particular interest to land managers in the Intermountain West of the USA. Currently, cold deserts of the Intermountain West are occupied with over 18 million ha (44,600,000 acres) of one or both *Juniperus* and *Pinus* species (Miller and

Tausch, 2000; Miller *et al.*, 2008). In general, the impact of hydrophobicity is scale dependent. At small scales, hydrophobic soils reduce soil wetting considerably and enhance runoff responses (Doerr *et al.*, 2000). On the other hand, rainfall simulation tests by Roundy *et al.* (1978) showed significantly greater infiltration capacity under well-developed *Juniperus* and *Pinus* mounds than in the intercanopy space. Because hydrophobicity can extend several centimetres into the soil profile (Jaramillo *et al.*, 2000), infiltration and redistribution of soil moisture throughout the soil matrix depends on wet patch conduits. Previous research suggests that uneven soil wetting associated with hydrophobic soil near established woody plants is a water conservation mechanism, as these surface hydrophobic zones would minimize evaporation losses from the soil surface (Scott, 1992; Moore and Blackwell, 1998; Jaramillo *et al.*, 2000).

In pinyon–juniper woodlands, prior field studies with various types of saturated or near saturated infiltration measurements have found increased infiltration under tree canopies relative to the intercanopy region (e.g. Blackburn and Skau, 1974; Wilcox *et al.*, 2003), with little evidence of how hydrophobicity affects infiltration. However, soil hydrophobicity decreases with increased soil moisture and cannot be accounted for in studies made at high soil moisture (Scholl, 1971; Doerr *et al.*,

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2000). Since many conventional infiltration and saturated hydraulic conductivity measurements are performed to achieve steady state flux at or near saturation, the control of surface hydrophobicity on infiltration response during the initial stages of an event is not characterized. This property is critical in many arid regions, where a significant portion of annual rainfall occurs as high intensity, short duration storms (Reid *et al.*, 1999). Furthermore, since hydrophobic soils may never reach saturated conditions throughout the soil matrix, and dry antecedent moisture conditions feedback to decreased soil moisture during precipitation events (Kobayashi and Shimizu, 2007), methods which measure hydraulic properties under saturated and/or steady state conditions may insufficiently characterize infiltration in arid rangeland ecosystems.

Understanding how woody plants modify the ecological processes within the intercanopy region is also critical in understanding the effects of pinyon–juniper establishment. One approach has focussed on dividing the landscape into distinct scale-dependent hydrologic functional units (HFUs) (Reynolds and Wu, 1999; Wilcox *et al.*, 2003) to better quantify the landscape patterns and processes (Ludwig *et al.*, 2005). Although this approach has merit, it assumes that ecohydrologic properties in the intercanopy are modified uniformly by the woody plants. However, the influence of woody vegetation on ecohydrologic properties may vary across a gradient with distance from the plant. For example, in an early study of rangeland ecohydrology, Seyfried and Wilcox (1995) hypothesized that improved infiltration capacity could extend beyond the canopy edge and thus have a significant role in patch to landscape scale infiltration processes (Ludwig *et al.*, 2005). Recent studies have shown woody plants can affect soil microclimate well into the intercanopy region through shading and interception, while affecting under ground resources through root uptake (Breshears, 2006; Lebron *et al.*, 2007). In a transect study with a *Pinus* tree and a *Juniperus* tree as endpoints, Lebron *et al.* (2007) attributed a gradient in soil moisture between the canopy and intercanopy HFUs to interception, and modified the existing HFU model to address meteorologically-driven directional gradients in ecohydrological processes. They observed a similar gradient in unsaturated hydraulic conductivity away from the *Juniperus* canopy into the intercanopy, but it was not clear whether the result was solely dependent on soil moisture.

In this paper, we assess whether infiltration in pinyon–juniper woodlands can be described as spatial gradients, which correspond to soil hydrophobicity and hydraulic conductivity as determined by tree position and species. This was accomplished by quantifying: (i) the ecohydrologic properties under the woody vegetation canopies of Utah juniper (*Juniperus osteosperma*) and pinyon pine (*Pinus edulis*); and (ii) how ecohydrologic properties changed with distance from the vegetation canopies and within the biological soil crust dominated intercanopy. Hundreds of ecohydrologic measurements were taken to characterize soil properties related to the major controls

on unsaturated infiltration, including hydrophobicity, biological soil cover, soil water content (SWC), short time infiltration (sorptivity), and steady state soil matrix infiltration (unsaturated hydraulic conductivity) at a high spatial resolution along transects extending from the base of *Pinus* and *Juniperus* trees to several meters within their respective intercanopy patch. These results may have important implications for surface hydrology, soil moisture dynamics, ecosystem function, and rangeland management.

MATERIALS AND METHODS

Study area

The study site is located on the Colorado Plateau near Moab, Utah, southwest of Dead Horse Point State Park, (38° 32'38.02 N, 109° 44'01.73 W), at an elevation of 1829 m. The climate is semi-arid, with a mean annual precipitation of 21.7 cm (Davidson *et al.*, 2002). The hydrology of this region appears more sensitive to the timing and characteristics of individual events than to seasonal trends in precipitation and temperature (Spence, 2001). Soils are developed from sandstone parent material, to form the Razono–Rock outcrop complex (loam, mixed (calcareous), mesic, Lithic Ustic Torriorthents) and are generally thin and poorly developed. Texture of the soils was found to be loamy sand with similar texture within both the subcanopy soil and intercanopy (Lebron *et al.*, 2007). Soil organic matter was higher under vascular plant canopies, where rich organic O and slight A1 horizons develop, than the intercanopy zone. The vegetation structure of the study location is described well by West (1999), i.e. mixed woodland of large *Pinus* and *Juniperus* trees, with the intercanopy soil surface covered with well-developed biological soil crust. An aerial view of the site and the study transects is shown in Figure 1. We found that cover from *Juniperus* and *Pinus* trees composed 26% of the site's land surface from analysis of colour aerial-photographs taken in the fall of 2006 (Utah Automated Geographic Reference Center) at 1-m resolution using Feature Analyst software extension (Visual Learning Systems Inc., Version 4.1) for ArcGIS, with density of tree and minimum distance between trees estimated at 111.7 trees/ha and 7.04 ± 0.10 m respectively, measured through Hawth's Tools extension for ArcGIS (Beyer, 2004).

Study design

Studies were conducted March 14–20, 2006, after a snowfall of approximately 13 cm. Within a 3 ha area, we chose seven *Juniperus* and *Pinus* trees that were of typical size and had various degrees of biological soil crust development within the intercanopy area (Figure 1). To assess the distribution of important hydrological parameters for soil wetness patterns developed under natural conditions, measurements were done just after the snow melted. The variable timing of melt due to surface shading by the canopies allowed us to make several replicate

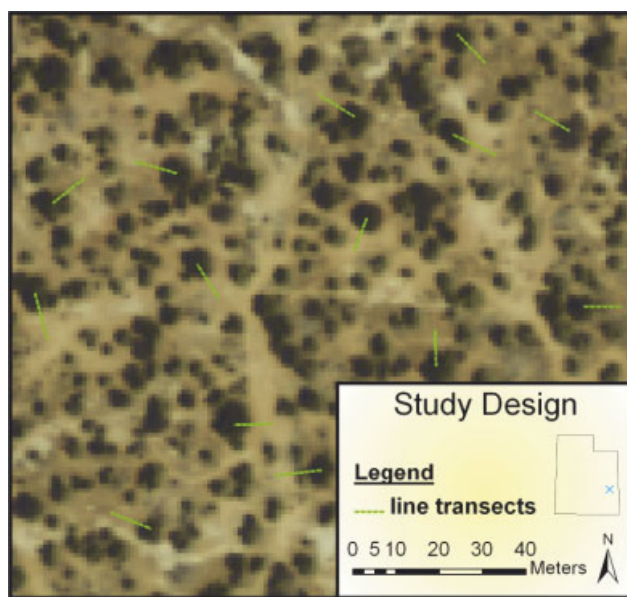


Figure 1. Aerial photograph of study site showing study trees as end-points for radial transect surveys. Site location in Utah, USA is inset.

measurements under similar conditions by following the melt throughout the site. For each tree we measured soil physical properties related to infiltration every 30 cm along one radial line transect, extending out from the base of the woody vegetation to approximately the centre of the intercanopy area. Transects were oriented to provide a reasonably even representation of cardinal orientation and to avoid intersection of obstacles such as fallen logs, disturbed areas, bare rock, and erosion features (Figure 1). Distance from the trunk to canopy edge was similar for *Juniperus* and *Pinus*, with a mean distance of 1.80 m (± 0.4 m). Distances from the base of the tree to approximately the centre of the intercanopy were also similar, with a mean distance of 4.45 m (± 0.65 m). Within the subcanopy, we repeated measurements of soil hydrologic properties following soil saturation and drainage for three of the seven *Juniperus* and *Pinus* line transects. This study design gave us a total of 274 sampling points, with 234 line transect measurement points within natural antecedent soil moisture conditions and 40 on subcanopy soil under pre-saturated conditions.

Measurements

At each measurement point along the radial line transects, soil sorptivity (S), unsaturated hydraulic conductivity [K(h)], and volumetric SWC were measured and biological soil crust cover was assessed. Sorptivity and K(h) were measured *in situ* with 15.9 mm Mini-disk Infiltrimeters (Decagon Devices Pullman, WA) at a head of -2.0 cm suction. Automation methods, measurement procedures, and calculations were performed according to Madsen and Chandler (2007). We chose to make measurements at -2 -cm head to maximize spatial replication, at the expense of characterizing K(h) at several head values. Use of the fixed head infiltrimeters also simplifies field logistics because they are relatively insensitive to

inclined deployment on sloping terrain. All measurements were corrected to standard temperature (20°C) by the viscosity ratio approach of Constantz (1982). Sorptivity was calculated from the first 30 ml drained from the infiltrimeter, excluding the initial 4 ml of water, which was disregarded to minimize the influence of contact sand on the result. Unsaturated hydraulic conductivity was calculated from infiltration at approximately steady state. Rosetta Lite (Schaap *et al.*, 2001) was also used to predict K(h) from soil texture. SWC was measured from the top 6 cm of soil with an ML2x Theta Probe (Delta-T Devices, Cambridge, England). Voltage from the Theta probe was converted to apparent permittivity and consecutively to SWC using the relationship given in Blonquist *et al.* (2005). Data were discarded from ten S, one K(h) and one SWC measurements due to measurement errors. At each transect point, categories of biological soil crust development were recorded as: (1) light crust (soil light in appearance, with a low biomass of cyanobacteria and low pinnacles); (2) dark crust (typically soil covered with cyanobacteria and lichens such as *Placidium spp.*, *Fulgensia spp.*, *Psora decipiens*, *Aspicilia spp.*, and *Toninia spp.* and *Psora cerebriformis*); and (3) moss crust (primarily *Syntrichia caninervis*) (Figure 2).

Within the subcanopy, tree litter depth was measured and then removed down to the mineral soil surface. Soil surface hydrophobicity was measured using the water drop penetration time test (WDPT) (Krammes and DeBano, 1965). With this same method, we also measured the depth of hydrophobicity in a pit-excavated midway between each *Juniperus* or *Pinus* trunk and canopy edge. We measured hydrophobicity at 1.0-cm depth increments until WDPT was less than 5 s. To force wetting of subcanopy soils for 'reference' infiltration measurements, we inserted 10-cm diameter, 15-cm deep sheet metal cylinders through the litter and slightly into the underlying mineral soil. The cylinders were filled with water and allowed to drain. This procedure was repeated until no hydrophobicity was detected.

A second survey of biological soil crust composition was undertaken to quantify our observation of a directional difference in the development of biological soil crust around trees. Observations of biological soil crust were made within 45° sections that covered a radial distance of 0.6 m, beginning at the canopy edge and extending 1.8 m into the intercanopy. Replicate surveys were performed on 12 individual *Juniperus* and *Pinus* trees.

Data analysis

As the total transect lengths were variable, measurements taken at 30 cm intervals are presented at their normalized distance with respect to the individual canopy. This was calculated as the ratio of the distance of the observation from the trunk to the distance of canopy edge from the trunk. Using these normalized canopy radius (CR) distances, data were grouped within quartiles (Q) at CR intervals of 0.25, to a distance of three times CR. For example, CR 0.50 is the point midway between the tree

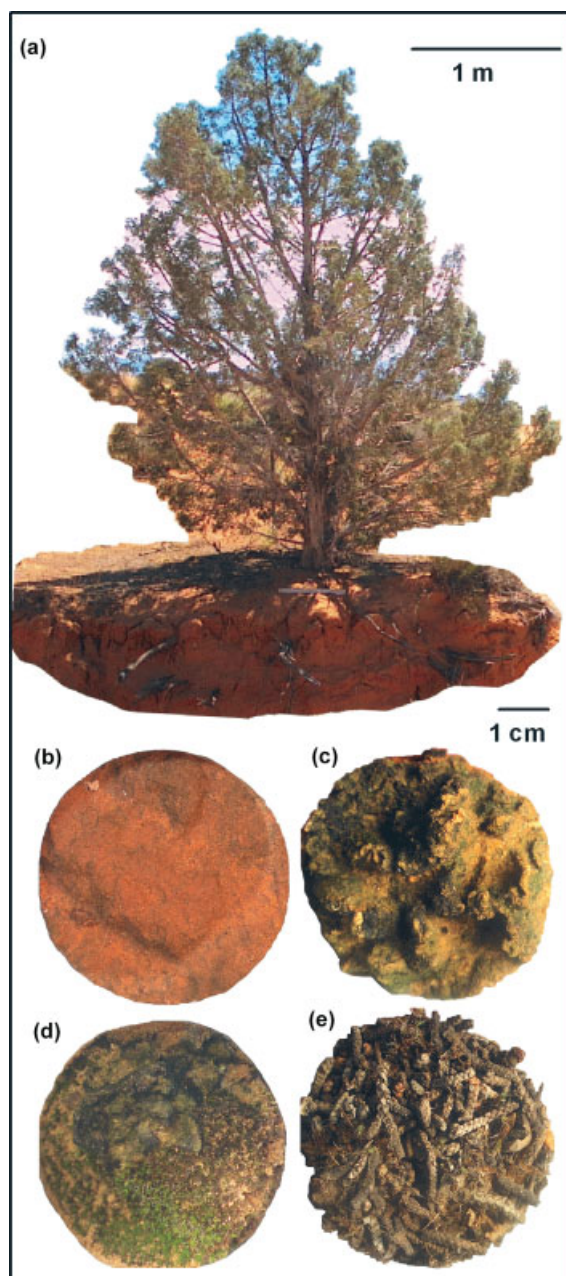


Figure 2. Photographs of (a) juniper pine tree (*J. sotosperma*), (b) light biological soil crust, (c) dark biological soil crust, (d) moss and lichens, (e) subcanopy litter.

trunk and the canopy edge and Q 0.50 is the spatial unit representing data points from CR 0.25 to 0.50. With both *Juniperus* and *Pinus* line transects combined, there is an average of 22 measurements per Q for dry antecedent soil moisture conditions and 9 measurements on hydrophobic soil, which was made wettable by saturating the soil until hydrophobicity was overcome.

Secondly, we analysed litter depth, biological soil crust composition, SWC, WDPT, $K(h)$ and S within patch and unit HFU scales with HFUs defined similar to Wilcox *et al.* (2003) as 'discrete and scale-dependent landscape units having hydrologic characteristics that are internally homogenous and quantitatively and qualitatively different from those of its immediate surroundings'. At the patch

scale, we separated the landscape into subcanopy and intercanopy HFUs at the canopy edge (CR 1.0), and also by alternate units that emerged from our measurements: CR 0–0.75, 0.75–1.75, and 1.75–3.0. At the unit scale, the subcanopy patch was grouped and analysed by tree species (*Pinus* or *Juniperus*) and the intercanopy patch measurements were grouped and analysed by biological soil crust development categories, using the groupings: light biological soil crust, dark biological soil crust, and moss biological soil crust.

Statistical analysis was performed with Sigma Stat 3.1 (Systat Software, Inc. Richmond, CA). For all comparisons a significance level of $P < 0.05$ was used. For each measurement type, cumulative data sets were assembled and tested for normality by Kolmogorov–Smirnov test. Litter depth and SWC were found to be abnormal and non-transformable through standard transformation functions. Therefore, we compared these data using non-parametric methods, and descriptive statistics were presented graphically through the median and 25th and 75th percentiles. For these parameters, differences between median Q values with distance were analysed through the Kruskal–Wallis ANOVA on ranks test, in association with the Dunn's test to determine significant differences between the variables. Both $K(h)$ and S were not normally distributed and were normalized by square root transformation prior to Q level comparison. Quartile mean values of $K(h)$ and S have been displayed graphically with standard error bars, with significance determined by a one-way ANOVA test and multiple comparisons among the different combinations of quartiles determined through the Holm–Sidak method. For comparison of the multiple intercanopy biological soil crust types, a one-way ANOVA test was also used in association with the Holm–Sidak method. Correlation among populations of $K(h)$, S , and SWC was analysed within larger CR segments at the patch and unit level by the Mann–Whitney rank sum test. Inter-comparison of SWC, litter depth, $K(h)$, and S was performed by CR segments by Spearman Rank correlation test, with results presented as correlation coefficients r and significant P values.

RESULTS

Soil surface cover

The soil surface was covered primarily with biological soil crust in the intercanopy and leaf litter under the plant canopy (Figure 2). In the intercanopy region, we found that while dark biological soil crust occurred at approximately 60% of the transect measurement locations beyond the canopy, moss occurrence declined from 20 to 10% between CR 1.5 and 2.5, with a complementary increase in light biological soil crust cover (Figure 3). A subsequent detailed survey of light and dark biological soil crust from a separate set of seven trees clarified the radial dependence of cyanobacterial biological soil crust development in this environment (Figure 4). Light biological soil crust occurred most frequently on the

southern side of the trees (from 100 to 200° from N) and dark biological soil crust occurred most frequently on the northern side of the trees ($0 \pm 50^\circ$ from N).

Median litter depths were similar for *Juniperus* and *Pinus* along the line transects. Leaf litter depth decreased radially from near the base of the tree (3–5 cm) to only a trace, within Q 1.00 (Figure 5(a)).

Soil water content

Upon arrival at the study site, we observed a uniform snow cover of approximately 13-cm depth in the intercanopy. The tree canopies had intercepted most of the snowfall to approximately CR 0.75, and snow depth was variable due to heterogeneity in canopy interception and surface shading. After snowmelt, we did not find significant differences between radial patterns or directionality in SWC among the *Pinus* and *Juniperus* transects, and thus combined these data for subsequent analysis (Figure 5(b)). At the patch HFU scale, the median intercanopy SWC ($0.19 \text{ m}^3 \text{ m}^{-3}$) was significantly greater

($P < 0.001$) than that of the subcanopy ($0.08 \text{ m}^3 \text{ m}^{-3}$). However, comparison of SWC by quartiles did not support a distinct boundary between HFU patches at the canopy edge (Figure 5(b)). In the subcanopy area, Q 0.25 and Q 0.50 showed the lowest median SWC values ($0.05 \text{ m}^3 \text{ m}^{-3}$), which are approximately one-quarter of the intercanopy median. A clear gradient in SWC emerges between Q 0.50 and Q 1.25, beyond which SWC remained similar, with median SWC ranging from 0.17 to $0.19 \text{ m}^3 \text{ m}^{-3}$.

In areas where the snow had been blown under the canopy and melted, the litter and duff material on the soil surface were completely saturated following snowmelt. However, the underlying soil was air-dry except for a few small wet patch conduits typically associated with fine roots growing close to the soil surface. From our observations, it appeared that the litter and duff materials detained water from running off the copious subcanopy mounds, while concentrating flow to the preferentially wetted flowpaths in the mineral soil.

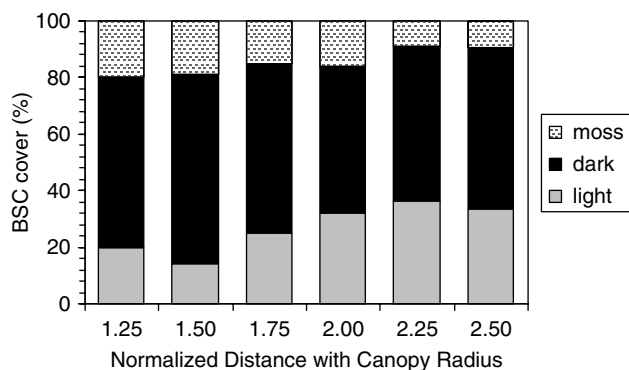


Figure 3. Percent cover of light, dark and moss biological soil crusts (BSC), within normalized distance quartiles.

Hydrophobicity

WDPTs of the soil at the interface of the mineral soil and organic material showed hydrophobicity extended from near the tree trunks to CR 0.75, on average, for both *Juniperus* and for *Pinus* trees (Figure 5(a)). Within this region, hydrophobicity at the interface between soil and litter material was so severe (WDPT > 15 min) that time constraints on the fieldwork precluded survey of soil hydrophobicity with depth at each sampling point. Water drop tests in excavations at CR 0.5 showed hydrophobic soils extended to as deep as 17.5 and 16 cm for *Juniperus* and *Pinus* respectively, with an average depth of 12.9 ± 1.4 cm for *Juniperus* and 10.0 ± 1.7 cm

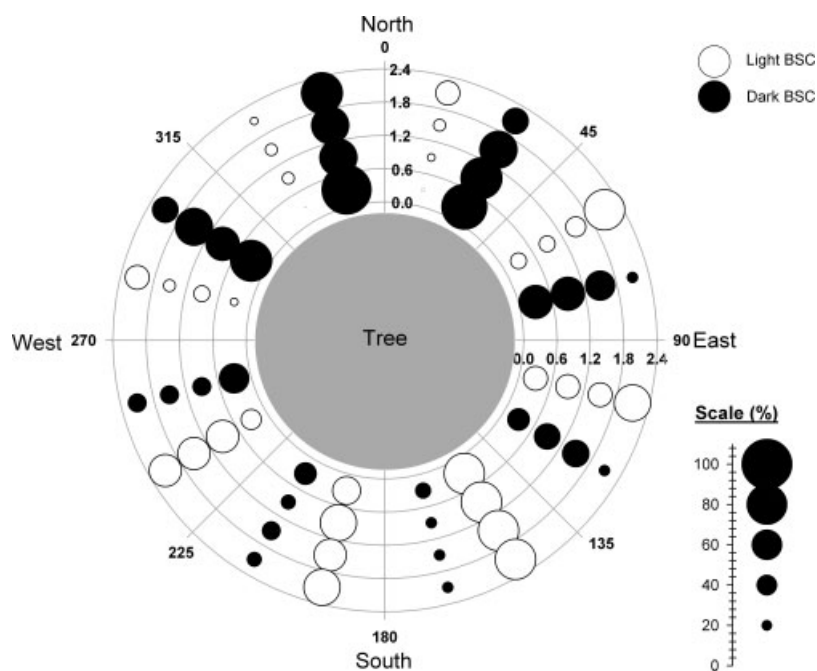


Figure 4. Spatial distribution of light and dark biological soil crust (BSC) by radial distance (in metres) and direction (degrees from North) away from tree canopies. Percentage cover for each category is represented graphically by the size of the dark circles.

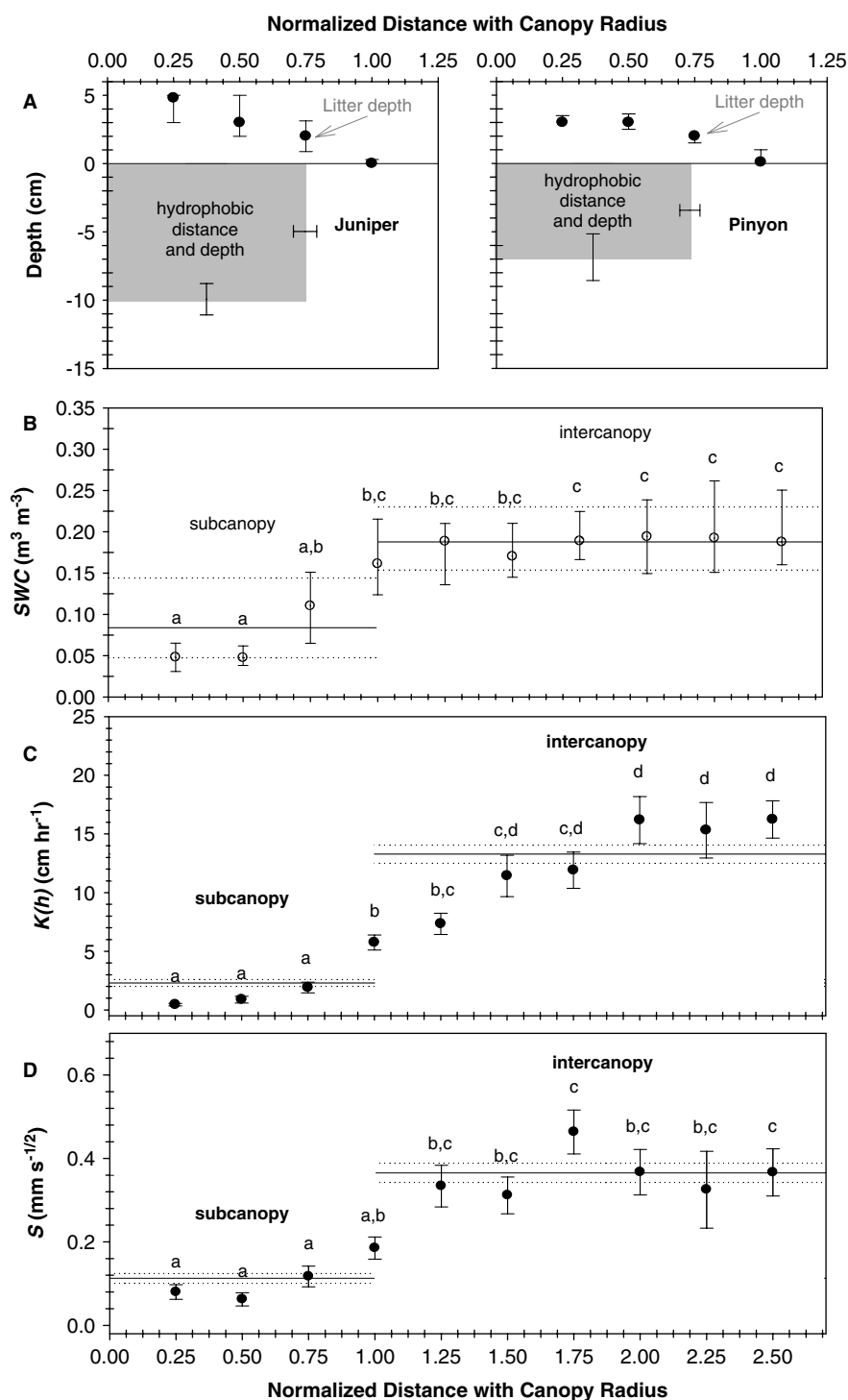


Figure 5. (a). Median litter depths (positive above soil surface, depth = 0) and associated 25th and 75th percentiles, for canopy radius quartiles. Soil water repellence is shown as surface extent from tree bole on the x-axis, and depth at 0.5, the canopy radius (CR) on the y-axis. (b) Soil water content (SWC) median, 25th and 75th percentiles by quartile (closed circles with error bars) and subcanopy and intercanopy patch scales (horizontal solid and dashed lines). Panels (c) and (d) present unsaturated hydraulic conductivity [$K(h)$] and sorptivity (S) in the same manner as panel (b). Statistical differences among quartile values are denoted by different letters.

for *Pinus*. Within Q 1.00, the incidence of surface hydrophobicity was variable, and in the intercanopy, the soil was not hydrophobic (Figure 5(a)).

Infiltration

Spatial gradients in both $K(h)$ and S blurred the lines between the subcanopy and intercanopy, with quartile level analysis results similar for both *Juniperus*

and *Pinus*. Figure 5(c) and (d) shows mean values and associated standard errors for $K(h)$ and S measurements presented at the Q and HFU scale, for combined data from *Pinus* and *Juniperus*. Quartile average values for $K(h)$ and S from Q 0.25 to 0.50 are less than the subcanopy patch average ($2.17 \pm 0.31 \text{ cm h}^{-1}$ and $0.11 \pm 0.01 \text{ mm s}^{-1/2}$, respectively). A positive gradient in $K(h)$ originated near the tree trunk, but is most clear

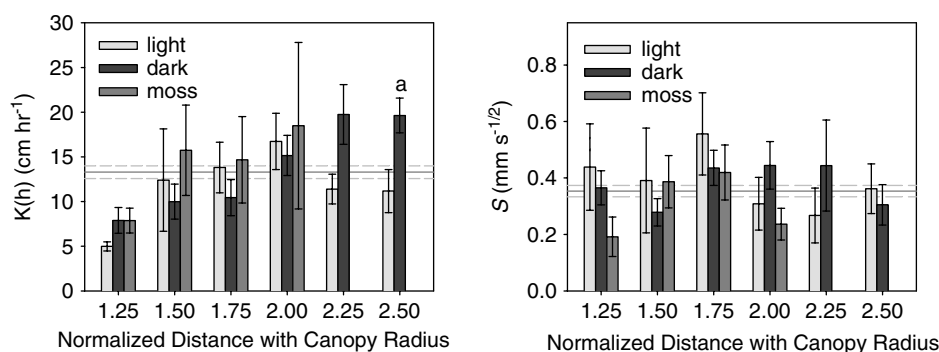


Figure 6. Unsaturated hydraulic conductivity [$K(h)$], and sorptivity (S) divided within the intercanopy by canopy radius distance quartiles and by light, dark, and moss biological soil crust types. Significant differences in $K(h)$ were found between light and dark BSC at CR 2.50 as denoted by lower case letter a. Horizontal solid and dashed lines represent intercanopy means and standard errors, respectively.

from Q 0.75 ($1.91 \pm 0.46 \text{ cm h}^{-1}$) to Q 2.0 ($16.18 \pm 2.01 \text{ cm h}^{-1}$) (Figure 5). Average $K(h)$ values for Q 1.00 ($5.75 \pm 0.64 \text{ cm h}^{-1}$) and Q 1.25 ($7.34 \pm 0.90 \text{ cm h}^{-1}$) are between the subcanopy and intercanopy ($13.29 \pm 0.79 \text{ cm h}^{-1}$) averages. Average $K(h)$ values from Q 2.0 and Q 2.5 (16.18 ± 2.01 and $16.23 \pm 1.59 \text{ cm h}^{-1}$, respectively) were significantly greater than the intercanopy average. For S , a shorter gradient emerged between the lowest value at Q 0.50 ($0.06 \text{ mm s}^{-1/2}$) and Q 1.25, beyond which, quartile values remained similar to the intercanopy average value ($0.37 \pm 0.02 \text{ mm s}^{-1/2}$) and ranged between 0.26 and $0.44 \text{ mm s}^{-1/2}$. The subcanopy gradients in S and $K(h)$ did not show a significant directional bias. However, the greatest values of $K(h)$ tend to occur on measurement transects between 225 and 45° from north and at CR greater than 1.75.

We tested the radial dependence of $K(h)$ and S on biological soil crust cover type within the intercanopy (Figure 6). For Q 1.25, mean $K(h)$ values for light biological soil crust, dark biological soil crust, and moss biological soil crust were all significantly lower than the intercanopy mean value (13.3 cm h^{-1}). Between CR 1.25 and 2.0, $K(h)$ increased by two to three-fold for dark biological soil crust (7.89 ± 1.45 – $15.14 \pm 2.24 \text{ cm h}^{-1}$) and light biological soil crust (4.98 ± 0.51 – $16.72 \pm 3.15 \text{ cm h}^{-1}$). Beyond CR 2.0, moss biological soil crusts were infrequent ($n = 2$ for each quartile) and thus were not included in the analysis. However, the difference in $K(h)$ between light and dark biological soil crust for Q 2.25 (11.39 ± 1.67 and $19.74 \pm 3.34 \text{ cm h}^{-1}$) and Q 2.50 (11.17 ± 2.41 and $19.62 \pm 1.95 \text{ cm h}^{-1}$) suggests that biological soil crust cover type influences $K(h)$ far from the trees. Furthermore, at this distance, $K(h)$ for dark biological soil crust was greater than both the intercanopy mean and modelled value (15 cm h^{-1}). Within the intercanopy, S did not vary systematically with either biological soil crust composition or distance from the canopy.

When hydrophobicity of the subcanopy soil was overcome by pre-wetting, $K(h)$ and S increased by several folds relative to the initial measurements (Table I). For example, within Q 0.25, $K(h)$ increased from 0.45 to

16.66 cm h^{-1} , and S increased from 0.08 to $0.47 \text{ mm s}^{-1/2}$ (Table I). The results of pre-wetting from near the canopy edge were less dramatic due to greater $K(h)$ under natural condition. For example, pre-wetted mean $K(h)$ for Q 1.0 (6.63 cm h^{-1}) was at least three times greater than Q 0.25, 0.50, and 0.75, respectively, but reached a similar rate following artificial wetting. On the other hand, the pre-wetted $K(h)$ values at Q 0.25 were greater than the average intercanopy $K(h)$ values; the remainder of the $K(h)$ and S values for pre-wetted subcanopy soils were not significantly different from snowmelt-wetted intercanopy soils beyond CR 2.0 and 1.25, respectively (Table II).

Hydrologic functional distributions

We separated the data into subcanopy and intercanopy (patch level) HFUs, at CR 1.0, and then into three HFU patches: the hydrophobic subcanopy (CR 0–0.75), the transitional gradient in $K(h)$ (CR 0.75–1.75), and the unaffected intercanopy past CR 1.75 (Table II). We tested for statistical differences in $K(h)$, S , and SWC at the unit level within each of these schemes and for the transect level (global) data. For the global data sets, SWC had a bimodal normal distribution, whereas $K(h)$ and S were better represented by a Poisson distribution. When these data were divided by HFU, S and SWC were normally distributed beyond CR 0.75, and all data were

Table I. Comparative values of unsaturated hydraulic conductivity [$K(h)$] and sorptivity (S) for measurements performed within the subcanopy with the soil hydrophobic and following artificial wetting. Descriptive statistics show mean and standard error, for measurements by quartiles of normalized distance from tree bole to canopy radius.

Canopy radius	$K(h)$ (cm h^{-1})				S ($\text{mm s}^{-1/2}$)			
	Hydrophobic		Pre-wetted		Hydrophobic		Pre-wetted	
	Mean	SE	Mean	SE	Mean	SE	Mean	SE
0.25	0.45	0.11	16.66	2.18	0.08	0.02	0.47	0.10
0.50	0.88	0.29	10.55	1.12	0.06	0.02	0.59	0.09
0.75	1.91	0.46	11.40	2.11	0.12	0.03	0.60	0.10
1.00	5.75	0.64	10.96	3.17	0.19	0.03	0.54	0.20

Table II. Descriptive statistics for unsaturated hydraulic conductivity [K(h)], sorptivity (S) and soil water content (SWC) measurements (SWC). Sample populations (n) were selected by subdividing the global data set (all) by normalized distance from the base of trees and by cover type (unit) or treatment (e.g. pre-wetted) within that distance. Median, mean, and standard error (SE) are presented, and normally distributed populations are indicated (*).

Normalized distance	Patch/Unit	K(h) (cm h ⁻¹)				S (mm s ^{-1/2}) × 10 ⁻¹				SWC (m ³ m ⁻³) × 10 ⁻¹						
		<i>n</i>	Median	Mean	<i>SE</i>	<i>n</i>	Median	Mean	<i>SE</i>	<i>n</i>	Median	Mean	<i>SE</i>			
0–3.0	Global	232	6.11	8.63	0.56	223	1.94	2.56	0.15	233	1.61	1.54	0.05			
0–1.0	Subcanopy	83	0.64	1.73	0.29	82	0.679	1.02	0.12	83	0.63	0.86	0.06			
	All pre-wet	40	17.12	18.44	1.68	*	40	4.67	5.64	0.63	*					
	Pinyon	42	0.69	1.66	0.34		41	0.647	0.87	0.14	*	42	0.94	0.94	0.09	
	Juniper	41	0.60	1.79	0.49		41	0.714	1.17	0.19		41	0.57	0.78	0.09	
0–0.75	Subcanopy	70	0.36	1.00	0.18		69	0.493	0.88	0.12		70	0.55	0.71	0.05	
	Pinyon	35	0.39	1.00	0.23		34	0.531	0.80	0.14	*	35	0.63	0.79	0.08	*
	Juniper	36	0.29	1.14	0.30		36	0.475	0.93	0.18		36	0.53	0.68	0.08	
1.0–3.0	Intercanopy	150	9.35	12.51	0.68		142	3.03	3.44	0.20	*	150	1.86	1.91	0.04	*
0.75–1.75	Transition	84	7.56	8.94	0.69		82	2.67	3.11	0.24	*	84	1.76	1.82	0.05	*
1.75–3.0	Intercanopy	79	13.34	15.18	1.00	*	73	3.38	3.52	0.29	*	79	1.93	1.98	0.06	*
	Light BSC	24	11.45	13.80	1.49	*	23	3.38	3.29	0.43	*	24	1.94	1.92	0.12	*
	Dark BSC	43	17.59	16.84	1.30	*	41	3.76	3.60	0.42	*	43	1.83	1.95	0.08	*
	Moss	12	7.80	12.00	3.51	*	9	2.68	3.76	0.96	*	12	2.44	2.19	0.14	*

normally distributed beyond CR 1.75. Subcanopy data were normally distributed only for pre-wetted conditions (Table II). To simplify comparison of normal and highly skewed populations, statistical tests of these data are presented for median values, and we also present mean and standard error as descriptive statistics (Table II). Within the subcanopy, median values of K(h), S, and SWC were not significantly different ($P = 0.14, 0.55, 0.17$, respectively) between CR 0–1.0 and CR 0–0.75. Similarly, unit level analysis comparing the differences between *Juniperus* and *Pinus* subcanopies shows they are statistically similar for SWC, K(h), and S, regardless of whether the subcanopy patch was defined as the CR 1.0 (P values = 0.21, 0.54, and 0.54 for SWC, K(h), and S, respectively) or CR 0.75 (P values = 0.21, 0.54, and 0.54 for SWC, K(h), and S, respectively). However, average K(h) values are significantly different between CR 0–1.0 and CR 0–0.75 (1.73 ± 0.29 and 1.00 ± 0.18 cm h⁻¹, respectively).

After pre-wetting to remove hydrophobicity, median subcanopy K(h) changed significantly from pre-wetted conditions ($P < 0.001$) and was significantly different from all HFU divisions except CR 1.75–3.0 ($P = 0.13$).

The populations of K(h), S, and SWC in the intercanopy and transition zones are all significantly different from the subcanopy ($P < 0.001$), but significant differences between the transition zone and intercanopy occur only for K(h). Further, K(h) is significantly different among CR 0.75–1.75 (8.9 ± 0.7 cm h⁻¹), CR 1.0–3.0 (12.5 ± 0.72 cm h⁻¹), CR 1.75–3.0 (15.22 ± 1.0 cm h⁻¹). Spearman rank order correlations among K(h), S, and SWC showed significant trends for CR 0–1.0 and CR 0–0.75 ($P < 0.004$), but no significant trends for CR 1.0–3.0, CR 0.75–1.75, and CR 1.75–3.0 (Table III).

These patterns in K(h), S, and SWC correspond with litter depth within the area of the subcanopy, where the soil is predominantly hydrophobic. Thus, it follows that Spearman correlation tests are negative between litter thickness in the subcanopy and K(h), S, and SWC (Table IV). However, we found S was not as strongly correlated to litter thickness as were K(h) and SWC at either the Q or single measurement level. This finding supports our field observation that variability in wetness of the soil O-horizon affected short time infiltration, S, independently of long time infiltration, K(h), which was more sensitive to the condition of the underlying mineral soil.

Table III. Spearman rank order correlation coefficient (r), P value, and number of samples (n) for comparisons among unsaturated hydraulic conductivity [K(h)], sorptivity (S), and soil water content (SWC) populations within patch level hydrological functional units as defined by different distances of canopy radius (CR) from the tree trunk. There are no significant relationships for $P > 0.050$.

		CR 0–1.0		CR 0–0.75		CR 1.0–3.0		CR 0.75–1.75		CR 1.75–3.0	
		S	SWC	S	SWC	S	SWC	S	SWC	S	SWC
K(h)	r	0.40	0.56	0.38	0.38	0.00	0.07	0.21	–0.04	–0.10	0.06
	P	<0.001	<0.001	0.001	0.001	0.993	0.410	0.065	0.687	0.419	0.570
	n	82	83	69	70	142	150	82	84	73	79
S	r		0.31		0.26		–0.01		0.06		–0.06
	P		0.004		0.029		0.901		0.590		0.590
	n		82		69		142		82		73

Table IV. Spearman rank order correlation coefficient (r), P value, and number of samples (n) for comparisons with litter depth to unsaturated hydraulic conductivity [$K(h)$], sorptivity (S), and soil water content (SWC). Significance was determined at $P < 0.050$.

		$K(h)$	S	SWC
Litter depth	r	-0.64	-0.26	-0.57
	P	<0.001	0.001	<0.001
	n	83	82	83

DISCUSSION

The most apparent influences of large woody vegetation on hydrology have previously been associated with the interception of precipitation and solar radiation (Breshears, 2006), enhanced infiltration under the canopy, and root uptake of soil water. (Seyfried, 1991; Pierson *et al.*, 1994; Wilcox *et al.*, 2003). Our finding of lower SWC in the subcanopy would seem to support previous findings of interception by *Pinus* and *Juniperus* trees (Owens *et al.*, 2006; Lebron *et al.*, 2007), although low soil moisture content is not a reliable indication of canopy interception in hydrophobic soils. On the other hand, measurements of lower $K(h)$ in the subcanopy when compared to the intercanopy would appear to contradict findings from previous infiltration studies in semi-arid rangeland environments. We contend that the difference in our findings is due to methodological differences which can be used to clarify the processes by which woody plants modify their environment through the development of litter mounds, hydrophobic soils, and roots, which we depict in Figure 7.

In semi-arid environments, two important features of the subcanopy environment of woody plants have not been well documented: extensive hydrophobicity of soils and moisture storage within the litter layer. When measured under ambient antecedent moisture conditions, severe hydrophobicity in Q 0.25, 0.50, and 0.75 (Figure 5) inhibited water entry into the soil fabric, with a

concomitant reduction in $K(h)$ relative to the intercanopy (Figure 5). Thus, assuming the presence of hydrophobicity, our results differed from those of Seyfried (1991) and Wilcox *et al.* (2003), who used measurement techniques that either removed or were not sensitive to soil hydrophobicity. When hydrophobicity was removed in our study by pre-wetting to saturation, subcanopy $K(h)$ was similar to the intercanopy $K(h)$, as found by Seyfried (1991) and Wilcox *et al.* (2003). We attribute the difference in pre-wetted $K(h)$ between Q 0.25 (16.66 cm h^{-1}) and the other pre-wetted subcanopy quartile averages (10.55 to 11.44 cm h^{-1}) (Table I) to modification of the soil structure by the rooting system of the plant or faunal activity under the canopy (Figure 7).

Our observations that litter mounds effectively retain moisture at near saturation supports the previous findings that there is lower runoff from the subcanopy than there is from the intercanopy (Pierson *et al.*, 1994), as runoff can be reduced by both storage at the soil surface and infiltration into the mineral soil. The additional observation of litter routing water preferentially to wetted spots in the underlying soil, and the frequent association of these wet spots with fine roots, further clarifies the apparent contradiction that despite extensive hydrophobicity, runoff is seldom observed from under *Pinus* and *Juniperus* canopies during precipitation inputs. We expect that low intensity precipitation inputs such as snow melt, or long duration, low intensity rainfall events, typical of winter frontal storms, may allow for the development of wet patches or complete removal of hydrophobicity. This would subsequently allow infiltration to occur at preferential locations under the canopy. However, high intensity monsoonal storms, typical of summer events, may deliver a sufficient depth of precipitation to exceed the storage capacity of the canopy and litter material. Further, precipitation at a rate greater than subcanopy soil S under hydrophobic conditions would conceivably generate runoff as observed by Reid *et al.* (1999) from runoff catchment plots.

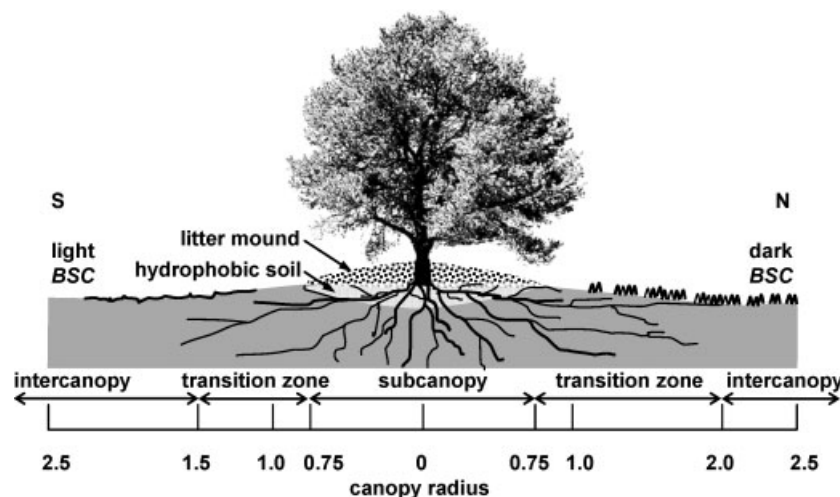


Figure 7. Schematic diagram of woody plant canopy and hypothetical root structure, litter mound and spatial distribution of dark and light biological soil crust (BSC), and how these relate to subcanopy, transitional zone, and intercanopy hydrological functional units.

In the intercanopy, we previously found that canopy interception of rain causes a transitional soil moisture zone east of *Juniperus* trees, due to prevailing winds from the west (Lebron *et al.*, 2007). However, in that study the radial measurements of $K(h)$ were only made in one direction, and the trends were complicated by the location of a heavy machinery track through the study site. In this study, snow deposition was relatively uniform in the interspace, and soil moisture did not vary significantly with distance from the canopy beyond CR 1.25. Although this pattern was reflected in the patterns of WDPT and S , a clear gradient in $K(h)$ emerged from within the subcanopy to approximately CR 2.0 in all directions (Figure 5). Beyond CR 0.75, $K(h)$ was not correlated with SWC or S (Table III). We conclude that the continual increase in $K(h)$ in all directions into the intercanopy indicates that the influence of woody vegetation on soil physical and chemical properties extends well beyond the canopy edge. In this study site, we suggest that sub-critical hydrophobicity from litter dispersal, runoff, and mycorrhizal root associations contributes to the observed gradient in $K(h)$ in the transitional zone into the intercanopy (Figure 7). Soil hydrophobicity develops as organic compounds on the surface of soil particles dry out, leaving a water-repellent coating surrounding the mineral soil (Doerr *et al.*, 2000). Wind-blown litter from the canopy and runoff from the elevated litter mounds into the intercanopy may result in lateral transport of hydrophobic compounds and subcanopy litter into the intercanopy area. However, we only observe a trace amount of litter beyond the subcanopy, and would expect this mechanism to develop a pattern related to the prevailing wind direction and existing stand structure. On the other hand, organic compounds in fine roots and associated mycorrhizae have previously been shown to result in hydrophobicity in dry soils (Hallett *et al.*, 2003). Our results agree with those of Hallett *et al.* (2003), who hypothesized that low levels of hydrophobicity would be responsible for high spatial variability in water transport, particularly under conditions of tension infiltration when macropores are inactive. While the extent of hydrophobic compounds within the transitional zone may not be great enough to limit soil wettability, the occurrence of organic coatings in the soil fabric may be sufficient to reduce infiltration in unsaturated conditions. We reason that the influence of hydrophobicity on $K(h)$ would decrease with distance from the canopy, regardless of whether the organic compounds originated from the canopy, roots, or associated mycorrhizae. Although spatially distributed data on root distributions for these species are difficult to obtain, as they can vary widely from site to site, we consider root related hydrophobicity and soil structural effects the likely causes of the observed transition zone gradient in $K(h)$ where hydrophobicity was not detected, due to the radial symmetry of the observations (CR 0.75–2.0 in Figure 5(c)).

In addition to the influence of litter and roots on soil physical properties, $K(h)$ may also be indirectly affected by the influence of the tree canopy on soil microclimate.

Pinyon–juniper woodlands often have wide interspaces among the trees, where the intercanopy soils experience more variable temperatures than subcanopy soils (Breshears *et al.*, 1998). Although the shelter effect of the canopy can vary due to factors such as leaf area, density, and height and spacing of the woody plants (Martens *et al.*, 2000; Breshears, 2006), the shelter provided by the canopy likely decreases from the tree trunk outwards to the canopy edge. Differences in the number of freeze-thaw cycles at the soil surface can also affect patterns of $K(h)$ by seasonally decreasing soil bulk density (Lampuranes and Cantero-Martinez, 2003). Since this study was conducted at the end of winter after the soil had experienced several freezing and thawing events, gradients in $K(h)$ could be due to variations in freeze-thaw effects on soil bulk density, especially within the regions beyond the reach of litter fall and roots (Figure 6). The difference in $K(h)$ beyond CR 1.75 may also be attributable to morphological variations among biological soil crust types, which is also associated with freeze-thaw cycles (Belnap, 2003).

We found light (least well-developed) biological soil crust mostly on the south of the trees, indicating that biological soil crust type is more strongly dependent on canopy interception of radiation than precipitation at this site (Figure 4). Similar to other infiltration studies (e.g. Eldridge *et al.*, 1997; Williams *et al.*, 1999; Wilcox *et al.*, 2003), $K(h)$ and S were not significantly different among biological soil crust types at the intercanopy HFU scale. However, near the canopy edge, $K(h)$ was significantly lower than the intercanopy average for all biological soil crust types and an increasing trend in $K(h)$ was observed between CR 1.0 and 2.0, with biological soil crust types similar within the same Q (Figure 6). From Q 2.0 to 2.5, $K(h)$ increased for dark biological soil crust and was almost double than that of light biological soil crust at Q 2.5 (light biological soil crust = 11.1 cm h^{-1} , dark biological soil crust = 19.6 cm h^{-1}). Coupling this result with the findings that well-developed biological soil crust occurs more frequently on the north side of the trees, may indicate that the woody plant canopy has a positive influence on $K(h)$ for CR 2.0–2.5 through its influence on biological soil crust development (e.g. Figure 7). Although the canopy also exerts a control on biological soil crust development from CR 1.0–2.0, we found no evidence that biological soil crust composition influenced $K(h)$ in this transitional zone between subcanopy and intercanopy HFUs.

The observed transformations may confer several advantages in water and nutrient capture, and retention to woody plants in resource-limited environments. Biota growing in sandy soils within dryland regions can be limited by high evaporation and low water holding capacity of the soil. It is generally believed that one of the reasons woody vegetation can persist in semi-arid systems is through their ability to extract deep soil moisture resources via tap roots. Woody plant roots have also been shown to transfer soil moisture deeper and broader within the soil profile through hydraulic redistribution. This can

result in water conservation by minimizing evaporation losses at the soil surface and reducing the rate of uptake (Ryel *et al.*, 2003). Our results support the view that within coarse textured soils, patchy hydrophobicity may be another means by which woody plants modify the environment, resulting in greater soil moisture availability. This patchiness can channel soil moisture through a few wet root-filled patches with greater infiltration capacity than the surrounding soil. The predominantly dry hydrophobic soil, with low S and an overlying mulch layer, would also restrict upward movement towards the soil surface, thus minimizing evaporative losses from under the canopy (Scott, 1992; Moore and Blackwell, 1998; Jaramillo *et al.*, 2000). Enhanced frequency and duration of litter mound wetness, due to hydrophobic subcanopy soil, may also accelerate litter decomposition and increase nutrient and carbon flow to underlying tree roots. Whereas, soil hydrophobicity may be a water conservation mechanism for established plant species, it may also impede germination and establishment of other species in the dry hydrophobic layer at the soil surface (Osborn *et al.*, 1967). Although the mechanism by which *Juniperus* and *Pinus* reduce K(h) beyond the canopy is not clear, the effect of increased canopy cover would be a progressive decrease in the average infiltration capacity of the intercanopy. This process may also be advantageous to the woody plants if it results in more focussed recharge to below ground soil moisture reserves.

CONCLUSIONS

Results from this study demonstrate the influence of *Juniperus* and *Pinus* vegetation in modifying soil hydrologic properties through the development of a hydrophobic layer within the subcanopy, with both species exhibiting similar hydrologic properties. For an established woody plant, hydrophobic soil may be ecological advantageous by channelling precipitation inputs deeper into the soil through a few selected wet patches, thus minimizing evaporative losses at the soil surface.

Hydrologic studies within pinyon–juniper woodlands typically analyse the landscape either by partitioning it into canopy and intercanopy patches or through random placement of points disregarding patch structure (e.g. Blackburn and Skau, 1974; Reid *et al.*, 1999; Wilcox *et al.*, 2003; Rau *et al.*, 2005; Pierson *et al.*, 2007). Our results demonstrate that infiltration is significantly different between subcanopy and intercanopy soil/cover type, which would support measuring landscape hydrologic properties by HFUs. However, this study further shows that whereas differences in soil state variables such as hydrophobicity may be discrete between the subcanopy and intercanopy, differences in K(h) are not discrete, but depend on the distance from the canopy edge. We suggest that changes in infiltration properties with distance from the canopy are the result of sub-critical water repellency, and/or diminishing influences of the canopy on the soils microclimate, which limited the degree of freeze-thaw to the soil surface. Both of these mechanisms may

have a significant influence on ecohydrologic properties, from vegetation establishment to runoff and erosion. For a more accurate assessment of infiltration properties in semi-arid ecosystems, distance from the base of the tree and canopy edge need to be considered as controlling length scales.

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